A Hypergraph-Native Message Passing Layer Based on Eidi-Otter Connectivity Patterns

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Guiding question and thesis context

Guiding question.

Can one define message passing on hypergraphs directly from hyperedge connectivity patterns (Eidi–Otter [2]), rather than by first reducing the hypergraph to a graph or to a general relational structure(Taha et al. [1])?

Goal of this talk

To present a minimal, mathematically sound candidate layer based on Eidi–Otter connectivity patterns.

Taha et al.: relationalization

Taha et al. [1] provide a general language for topological message passing through relational structures:

$$\mathcal{R} = (S, R_1, \ldots, R_k),$$

where S is a finite set of entities and each $R_i \subseteq S^{n_i}$ is a relation of arity n_i.

- Simplices, cells, or other higher-order objects become entities.
- Boundary, co-boundary, lower, and upper adjacencies become relations.
- Shift operators encode the strength of message passing through each relation.
- Aggregating these operators gives influence graphs, which allow graph-theoretic analysis.

Why this matters here

This is a powerful framework, but my construction asks for a more direct hypergraph route: start from incidence and hyperedge connectivity patterns themselves.

Hypergraphs and clique expansion

Hypergraph. A finite undirected hypergraph is

$$\mathcal{H} = (V, \mathcal{E}), \qquad \mathcal{E} \subseteq 2^V \setminus \{\emptyset\}.$$

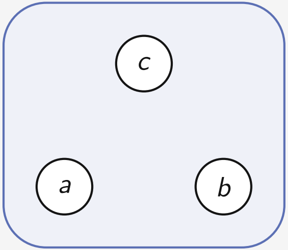
For two vertices $v, u \in V$, define the shared hyperedges

$$\mathcal{E}(v, u) = \{e \in \mathcal{E} : v \in e,\ u \in e\}.$$

Clique expansion. A hyperedge $e = \{v_1, \ldots, v_k\}$ is replaced by the complete graph on its vertices:

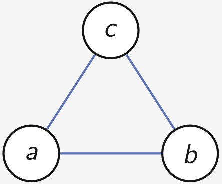
$$e = \{v_1, \ldots, v_k\} \quad \longmapsto \quad \{\{v_i, v_j\} : i \neq j\}.$$

Clique expansion can erase higher-order structure



$\mathcal{H}_1 : \mathcal{E}_1 = \{\{a, b, c\}\}$

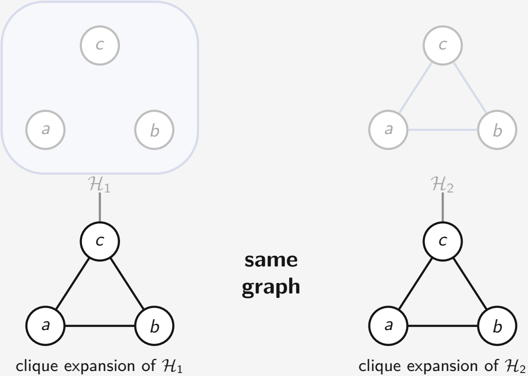
one genuine 3-way interaction



$\mathcal{H}_2 : \mathcal{E}_2 = \{\{a, b\}, \{a, c\}, \{b, c\}\}$

three separate pairwise interactions

Clique expansion can erase higher-order structure



Consequence

A purely graph-based GNN may erase the difference between one genuine 3-way interaction and three separate pairwise interactions.

Eidi–Otter: the ingredient I need

Eidi–Otter [2] study structural and regular equivalences in graphs and hypergraphs.

For this construction, the essential point is:

Key idea

Hypergraph geometry can be defined directly from how vertices are connected through hyperedges, and not only from the pairwise graph projection.

The specific ingredient used here is the **equal-edges random walk** from Eidi–Otter [2].

It is useful because it is sensitive to:

- Which hyperedges contain both vertices.
- The cardinalities of these hyperedges.
- The local hypergraph degree of the source vertex.

The equal-edges random walk

For a vertex v, define

$$\mathcal{E}(v) = \{e \in \mathcal{E} : v \in e,\ |e| \geq 2\}, \qquad d_{\mathcal{H}}(v) = |\mathcal{E}(v)|.$$

The equal-edges walk from v works as follows:

1. Choose uniformly one incident hyperedge $e \in \mathcal{E}(v)$;
2. Choose uniformly one vertex $u \in e \setminus \{v\}$.

Therefore, for $u \neq v$,

$$P_{\mathrm{EE}}(v, u) = \frac{1}{d_{\mathcal{H}}(v)} \sum_{e \in \mathcal{E}(v,u)} \frac{1}{|e| - 1}.$$

Interpretation

A neighbour reached through a smaller hyperedge receives more weight than a neighbour reached through a larger hyperedge.

P_{EE} is more than clique adjacency

Clique expansion only remembers whether two vertices co-occur:

$$v \sim u \iff \exists e \in \mathcal{E} \text{ such that } v, u \in e.$$

The equal-edges walk remembers more:

$$P_{\mathrm{EE}}(v, u) = \frac{1}{d_{\mathcal{H}}(v)} \sum_{e \in \mathcal{E}(v,u)} \frac{1}{|e| - 1}.$$

So the transition depends on:

- How many hyperedges v and u share.
- The cardinalities of those hyperedges.
- The number of hyperedges incident to v.

This is the first hypergraph-native ingredient of the proposed layer.

Message passing: the only template we need

A graph MPNN [3] updates node representations by

$$m_v^{(\ell+1)} = \mathrm{AGG}\left(\left\{\mathrm{MSG}^{(\ell)}\left(h_v^{(\ell)}, h_u^{(\ell)}, e_{uv}\right) : u \in \mathcal{N}(v)\right\}\right),$$

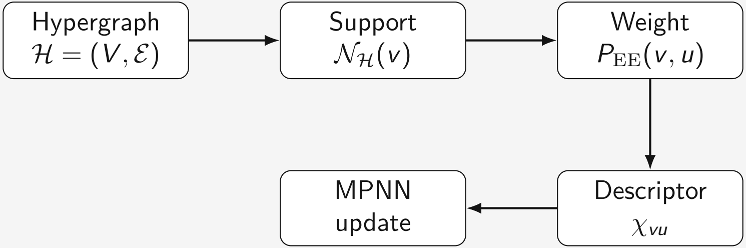
$$h_v^{(\ell+1)} = \mathrm{UPD}^{(\ell)}\left(h_v^{(\ell)}, m_v^{(\ell+1)}\right).$$

For this project, the important design choices are:

- The support of aggregation: who sends messages.
- The propagation weights: how strongly information moves.
- The message type: what structural information conditions the message.

Construction overview

I want to keep the MPNN structure, but replace graph-based propagation by hypergraph-derived quantities.



The support is vertex-to-vertex:

$$\mathcal{N}_{\mathcal{H}}(v) = \{u \in V \setminus \{v\} : \mathcal{E}(v, u) \neq \emptyset\}.$$

But the weights and message types are computed from the original hypergraph

Cardinality-pattern descriptor

Let $a_r \in \mathbb{R}^p$ be a learnable embedding associated with hyperedge cardinality $r \geq 2$.

For v, u with $\mathcal{E}(v, u) \neq \emptyset$, define

$$Z_{vu} = \sum_{e \in \mathcal{E}(v,u)} \frac{1}{|e| - 1}.$$

Then define

$$\chi_{vu} = \frac{1}{Z_{vu}} \sum_{e \in \mathcal{E}(v,u)} \frac{1}{|e| - 1} a_{|e|}.$$

Meaning

χ_{vu} is a learnable summary of the cardinalities of the hyperedges through which v and u interact.

Same clique, different descriptor

Consider again

$$\mathcal{H}_1 : \mathcal{E}_1 = \{\{a, b, c\}\}, \qquad \mathcal{H}_2 : \mathcal{E}_2 = \{\{a, b\}, \{a, c\}, \{b, c\}\}.$$

Both induce the same clique expansion.

But for the pair (a, b):

$$\chi_{ab}^{\mathcal{H}_1} = a_3, \qquad \chi_{ab}^{\mathcal{H}_2} = a_2.$$

Consequence

The proposed layer can distinguish some hypergraphs that are identical after unweighted clique expansion.

The EO-pattern message passing layer

At layer ℓ, each vertex has a representation $h_v^{(\ell)}$.

For each $u \in \mathcal{N}_{\mathcal{H}}(v)$, the message is conditioned on the cardinality pattern:

$$\psi^{(\ell)}\left(h_v^{(\ell)}, h_u^{(\ell)}, \chi_{vu}\right).$$

The aggregated message is

$$m_v^{(\ell+1)} = \sum_{u \in \mathcal{N}_{\mathcal{H}}(v)} P_{\mathrm{EE}}(v,u)\psi^{(\ell)}\left(h_v^{(\ell)}, h_u^{(\ell)}, \chi_{vu}\right).$$

The update is

$$h_v^{(\ell+1)} = \phi^{(\ell)}\left(h_v^{(\ell)}, m_v^{(\ell+1)}\right).$$

Relation to AllSet

AllSet [4] is a strong hypergraph-native baseline: it propagates through hyperedges using two learned multiset functions

$$V \longrightarrow E \longrightarrow V.$$

	AllSet	Proposed layer
Main idea	Learn multiset maps	Use EO geometry
Propagation	$V \to E \to V$	Vertex-centered EE walk
Native sense	Incidence-native	Geometry-native
Role here	Strong baseline	Interpretable construction

Positioning

AllSet is more general and probably more expressive. My layer is more constrained, but directly tied to Eidi–Otter connectivity patterns.

Properties and scope of the construction

Permutation equivariance.
Under any relabeling π,

$$\mathcal{E}(v,u) \mapsto \mathcal{E}(\pi v, \pi u), \qquad |e| \mapsto |\pi e| = |e|.$$

Thus $P_{\mathrm{EE}}(v,u)$ (weights) and χ_{vu} (descriptors) are preserved up to relabeling. Since $\psi^{(\ell)}$ and $\phi^{(\ell)}$ are shared across vertices and aggregation is a sum,

$$F(\pi\mathcal{H}, \pi X) = \pi F(\mathcal{H}, X).$$

Scope of the claim.

- The current contribution is a minimal geometry-driven construction.
- The construction preserves some cardinality information lost by unweighted clique expansion.
- Its empirical value and distinguishing behaviour will be assessed experimentally.

Guiding question

Guiding question.
Can one define message passing on hypergraphs directly from hyperedge connectivity patterns, rather than by first reducing the hypergraph to a graph or to a general relational structure?

Answer proposed in this talk

Yes, in a limited but precise sense: I propose a vertex-centered message passing layer whose weights and message descriptors are computed directly from hypergraph incidence and hyperedge cardinalities.

Thesis context

Project trajectory.

1. Survey of higher-order structures and message passing.
2. Survey of random walks and higher-order propagation.
3. Current step: propose a concrete hypergraph-native layer.
4. Next step: implement it and compare it experimentally.

Experimental plan

The next step is a minimal, controlled comparison.

Core baselines.

- Clique-expansion baselines: GCN [5], GIN [6].
- Uniform hypergraph MPNN with the same architecture.
- AllDeepSets / AllSetTransformer if integration is feasible [4].
- Proposed EO-pattern layer.

Ablations and checks.

- P_{EE} only.
- χ_{vu} only.
- Full model: $P_{\mathrm{EE}} + \chi_{vu}$.
- Synthetic tests where clique expansion erases cardinality information.

Main takeaway

The proposed layer is a minimal bridge between
Eidi–Otter hypergraph geometry
and the MPNN paradigm.

Core message

Hypergraph message passing can be defined directly from hyperedge connectivity patterns, at least in a vertex-centered and cardinality-sensitive form.

References

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Beyond the TFG

The TFG will focus on validating the minimal construction, not on building a full benchmark suite.

- Multi-step Eidi–Otter walks instead of only one-step EE transitions.
- Curvature-based gates using Ollivier–Ricci-type quantities.
- Richer connectivity descriptors beyond cardinality averages.
- Formal expressivity analysis: which hypergraphs can this family distinguish?
- Direct comparison with relational message passing and influence-graph constructions.

Long-term question

Can Eidi–Otter connectivity patterns define a broader family of hypergraph-native message passing architectures?

Backup: relational influence graphs

Given a relational structure

$$\mathcal{R} = (S, R_1, \ldots, R_k),$$

Taha et al. [1] associate shift operators to the relations and aggregate them into matrices

$$\widetilde{A} = \sum_{i=1}^{k} \widetilde{A}_{R_i}, \qquad B = \gamma I + \widetilde{A}.$$

The influence graph $G(S, B)$ has:

- entities S as nodes;
- an edge $\tau \to \sigma$ whenever $B_{\sigma,\tau} > 0$;
- edge weight $B_{\sigma,\tau}$.

This is the relational route: first encode the higher-order object as relations, then study message passing through an induced graph.

Backup: complete EO-pattern layer

Given $\mathcal{H} = (V, \mathcal{E})$, define

$$\mathcal{E}(v,u) = \{e \in \mathcal{E} : v, u \in e\}, \qquad \mathcal{N}_{\mathcal{H}}(v) = \{u \neq v : \mathcal{E}(v,u) \neq \emptyset\}.$$

$$P_{\mathrm{EE}}(v,u) = \frac{1}{d_{\mathcal{H}}(v)} \sum_{e \in \mathcal{E}(v,u)} \frac{1}{|e|-1}.$$

$$\chi_{vu} = \frac{1}{Z_{vu}} \sum_{e \in \mathcal{E}(v,u)} \frac{1}{|e|-1} a_{|e|}, \qquad Z_{vu} = \sum_{e \in \mathcal{E}(v,u)} \frac{1}{|e|-1}.$$

$$m_v^{(\ell+1)} = \sum_{u \in \mathcal{N}_{\mathcal{H}}(v)} P_{\mathrm{EE}}(v,u) \psi^{(\ell)} \left(h_v^{(\ell)}, h_u^{(\ell)}, \chi_{vu} \right).$$

Backup: limitations

- χ_{vu} is not a complete invariant of shared hyperedge structure.
- Different multisets of shared hyperedges may produce the same weighted average.
- The model is still vertex-centered.
- It does not maintain explicit hyperedge states.
- Empirical validation is still missing.

How to frame this

These are not fatal problems. They delimit the contribution: a minimal, interpretable, geometry-driven hypergraph layer.